# Ritesh Somashekar

**Data Analyst | SQL, Python & AI-Powered Analytics**

Fairfax, VA | (571) 245-7770 | ritesh.chandra33@gmail.com | LinkedIn

## PROFESSIONAL SUMMARY

- **Data Analyst** with experience collecting, cleaning, and transforming large, complex datasets into decision-ready insights across technology, healthcare, academic research, and industrial domains. Strong in **Python, SQL, R, Pandas, Spark,** and **statistical analysis**.
- Skilled in integrating disparate sources, establishing **data quality standards**, measuring KPIs, testing hypotheses, identifying trends and anomalies, and evaluating models through statistical metrics and **A/B testing**. Built reproducible **ETL and analytics workflows** on AWS and Azure.
- Delivers **reports, dashboards, and presentations** in **Tableau** and **Power BI**, translating findings into clear recommendations for stakeholders and senior management, and applies **machine learning, NLP, LLMs, and RAG** to analytical workflows.

## TECHNICAL SKILLS

**Data Analytics & Programming: SQL (PostgreSQL, Oracle)**, **Python**, **R**, Pandas, NumPy, Scikit-learn, Exploratory Data Analysis, Data Cleaning, Data Wrangling, Statistical Analysis, Trend Analysis

**Data Integration & Quality:** ETL/ELT, Apache Spark, PySpark, Airflow, Kafka, Delta Lake, Data Modeling, **Data Quality Standards**, Data Validation, Data Warehousing, Snowflake

**Visualization & Reporting: Power BI**, **Tableau**, Excel, Dashboard Development, Business Reporting, Executive Presentations, Data Storytelling

**AI/ML & Generative AI:** Machine Learning, NLP, BERT, BioBERT, Llama 3, GPT-4, Hugging Face, **RAG**, Vector Embeddings, FAISS, LightGBM, XGBoost, PyTorch, TensorFlow

**Experimentation & Platforms:** Hypothesis Testing, A/B Testing, KPI Analysis, Anomaly Detection, MLflow, Git, GitHub, DVC, AWS (S3, Glue, SageMaker, EMR), Azure (Synapse, Databricks), FastAPI, REST APIs, Docker

## EDUCATION

**Master's in Data Analytics and Engineering** | George Mason University, Fairfax, VA, USA

**Bachelor's in Information Science and Engineering** | Visvesvaraya Technological University, Bengaluru, Karnataka, India

## PROFESSIONAL EXPERIENCE

**George Mason University** | ***Graduate Research Assistant (Applied ML)*** **Fairfax, VA | August 2024 - Present**

- Developed NLP information extraction pipelines using **Python**, BioBERT, and Hugging Face to process **30,000 academic** documents, improving entity recognition **F1-score by 18%** over spaCy baselines.
- Collected, cleaned, and standardized **15GB** of structured and unstructured data across six datasets using Pandas and Scikit-learn, defining validation checks that reduced preparation time by **30%**.
- Analyzed large-scale disaster datasets with **Python, R,** and **Tableau**, applying statistical and ensemble models to forecast community coping trends with an **RMSE of 0.89** and presenting findings to research stakeholders.
- Performed feature engineering, model comparisons, and **statistical evaluation** to identify trends and patterns and assess the reliability of machine learning outputs for research applications.
- Built reproducible **Python** and MLflow experimentation pipelines with Optuna, reducing model iteration cycles from three days to **under 18 hours**.
- Deployed analytical inference APIs using **FastAPI and Docker** with **sub-200ms response latency**, supporting a multi-user research dashboard.
- Collaborated with research teams using Git and DVC to document analysis methods, track experiments, and maintain consistent evaluation benchmarks and reproducible workflows.
- Designed large-scale analytical and feature engineering pipelines using **Python**, Apache Spark, and **AWS EMR**, processing behavioral and telemetry data to support model development, performance tracking, and insights.
- Developed anomaly detection workflows using LightGBM, statistical evaluation, and threshold-based alerts for transaction monitoring, reducing false-positive alerts by **25%**.
- Evaluated AI/ML performance through **A/B testing**, accuracy metrics, and operational monitoring, helping teams compare models and make evidence-based deployment decisions.

- Automated analytical and model evaluation workflows using **MLflow and FastAPI**, reducing model release cycles from two weeks to five days and improving experiment reproducibility.
- Applied SHAP explainability to investigate feature contributions and prediction patterns, supporting model governance, risk assessment, and transparent analysis.
- Optimized Llama 3 models using LoRA and quantization-aware training, reducing model size by **40%** while keeping accuracy within 3% of full-precision benchmarks.

**Carelon Global Solutions** | ***Machine Learning Engineer*** **India | September 2020 - July 2023**

- Integrated structured and unstructured healthcare data with **PySpark, AWS Glue, S3, and Delta Lake**, building ETL pipelines that improved data reliability by **30%** under HIPAA requirements.
- Identified and resolved provider data accuracy issues by automating daily **NPPES** ingestion with **Python** and **SQL**, improving data accuracy by **74%** and cutting processing overhead by **60%**.
- Delivered self-service **Tableau** and **Power BI** reports backed by SQL data models, reducing reporting latency by **40%** for business stakeholders.
- Converted unstructured clinical text into analyzable entities using BERT and BioBERT, achieving **80% extraction accuracy** and reducing document processing time by **55%**.
- Performed statistical model evaluation, cross-validation, and hyperparameter tuning using Scikit-learn, XGBoost, and TensorFlow, increasing accuracy by **8%** and reducing training time by **15%**.
- Automated data ingestion, preprocessing, feature engineering, and recurring workflows through **Apache Airflow DAGs**, reducing manual processing effort by **25%**.
- Implemented **data quality checks** and validation standards across healthcare data workflows, supporting downstream analysis and real-time EHR integration through REST APIs and containerized services.
- Partnered with data engineering, product, and client stakeholders to define analytical KPIs, evaluate solution performance, and prioritize delivery, reducing turnaround time by **15%**.